# Introduction to Machine Learning in Epidemiology with R

From epidemiological questions to prediction, evaluation and interpretation



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# 01 What machine learning means in an epidemiological context

**Machine learning asks whether patterns in observed data can make useful predictions for new observations.**

**In epidemiology, the workflow must also respect:**

- The population and setting that generated the data
- When predictors and outcomes become available
- Surveillance, selection and measurement processes
- The consequences of false positives and false negatives

**Prediction quality depends on both the model and the data-generating process.**

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# From an epidemiological question to a prediction task

1. **Question**
   What decision could a prediction support?
2. **Unit**
   Who or what receives one prediction?
3. **Outcome**
   What event and prediction horizon matter?
4. **Predictors**
   What information exists at prediction time?
5. **Evaluation**
   Which errors matter in practice?

**Illustrative workshop framing**

**Predict whether reported activity will be elevated in a future period, using only information available before that period begins.**

03

# Preparing data for machine learning

## Keep time honest

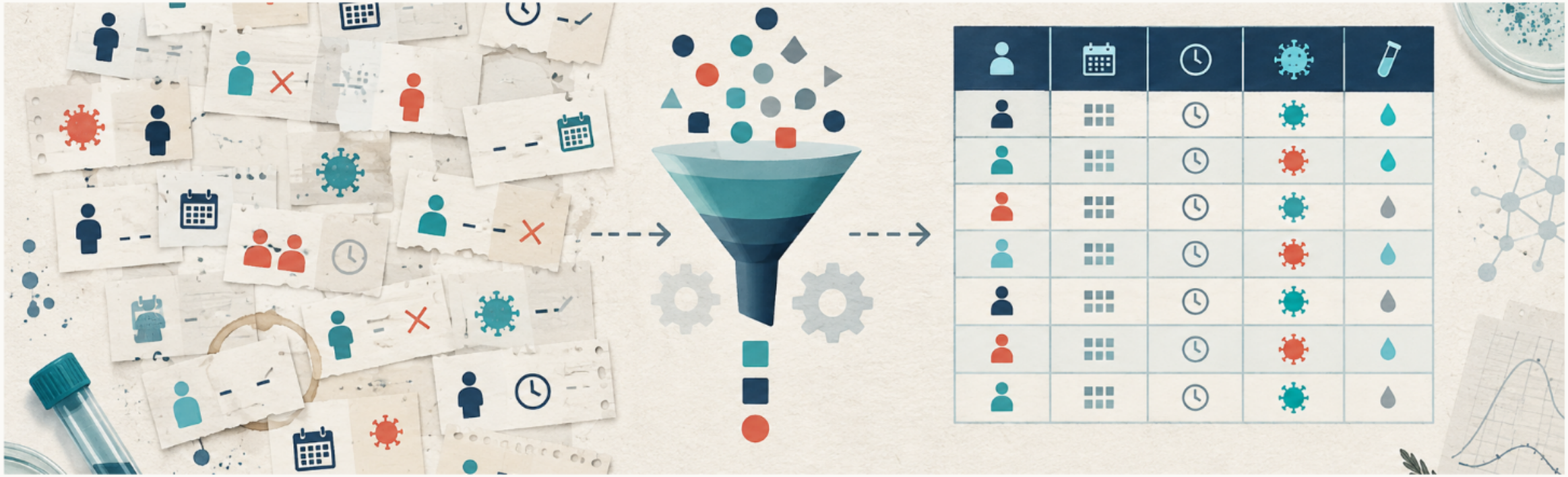
Create predictors from information already observed.

## Match the unit

One row should correspond to one prediction unit.

## Build inside resampling

Estimate preprocessing steps from training folds only.



# Logistic regression as a baseline model

## A baseline gives the more flexible models something meaningful to beat.

```
learner_log = lrn("classif.log_reg",
  predict_type = "prob")

learner_log$train(task, row_ids =
train_ids)
pred_log = learner_log$predict(task,
row_ids = test_ids)
```

### Why start here

- Produces predicted probabilities for a binary outcome
- Provides a familiar, interpretable reference point
- Reveals whether added flexibility improves unseen-data performance

**The comparison concerns prediction. Coefficients do not become causal effects automatically.**

# Decision trees and random forests

## Increasing flexibility changes both what the model can learn and how easily we can explain it.

| Question | Decision tree | Random forest |
|---|---|---|
| **How it learns** | Recursive splits form one tree | Many trees learn from varied samples and features |
| **Main strength** | Easy to inspect and explain | Often improves prediction and stability |
| **Main risk** | Can change sharply with small data changes | Individual prediction logic becomes less transparent |
| **mlr3 learner** | classif.rpart | classif.ranger |

**Model complexity earns its place only when performance improves on data the model has not seen.**

# Training and evaluating models with mlr3




```
task = as_task_classif(dat, target = "outcome", positive = "yes")
learner = lrn("classif.ranger", predict_type = "prob")
rr = resample(task, learner, rsmp("cv", folds = 5))
rr$aggregate(msrs(c("classif.auc", "classif.sensitivity",
  "classif.specificity")))
```

**The same task, folds and measures make model comparisons reproducible.**

# Cross-validation and performance on unseen data

## Five-fold cross-validation rotates the test fold so every observation is evaluated once.




**Resampling estimates generalisation performance. It does not remove overfitting or repair a flawed split strategy.**

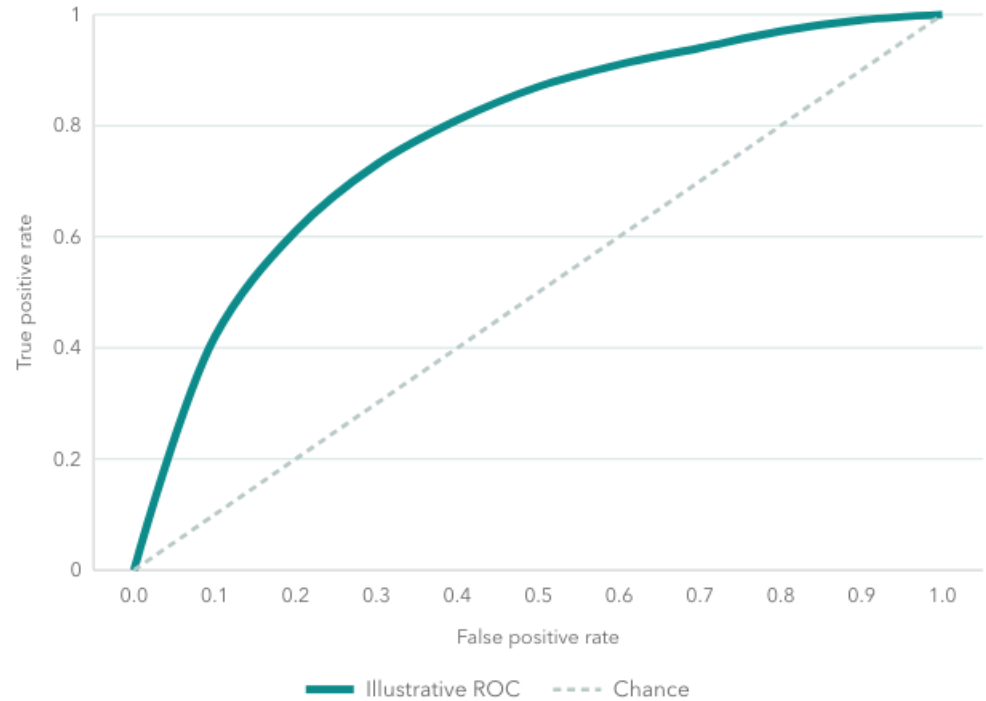
For time series, repeated geography or clustered data, use folds that preserve those structures.

# Sensitivity, specificity, confusion matrices and ROC/AUC

**Illustrative values**

| | Predicted + | Predicted – |
|---|---|---|
| Observed + | 32 | 8 |
| Observed – | 12 | 48 |

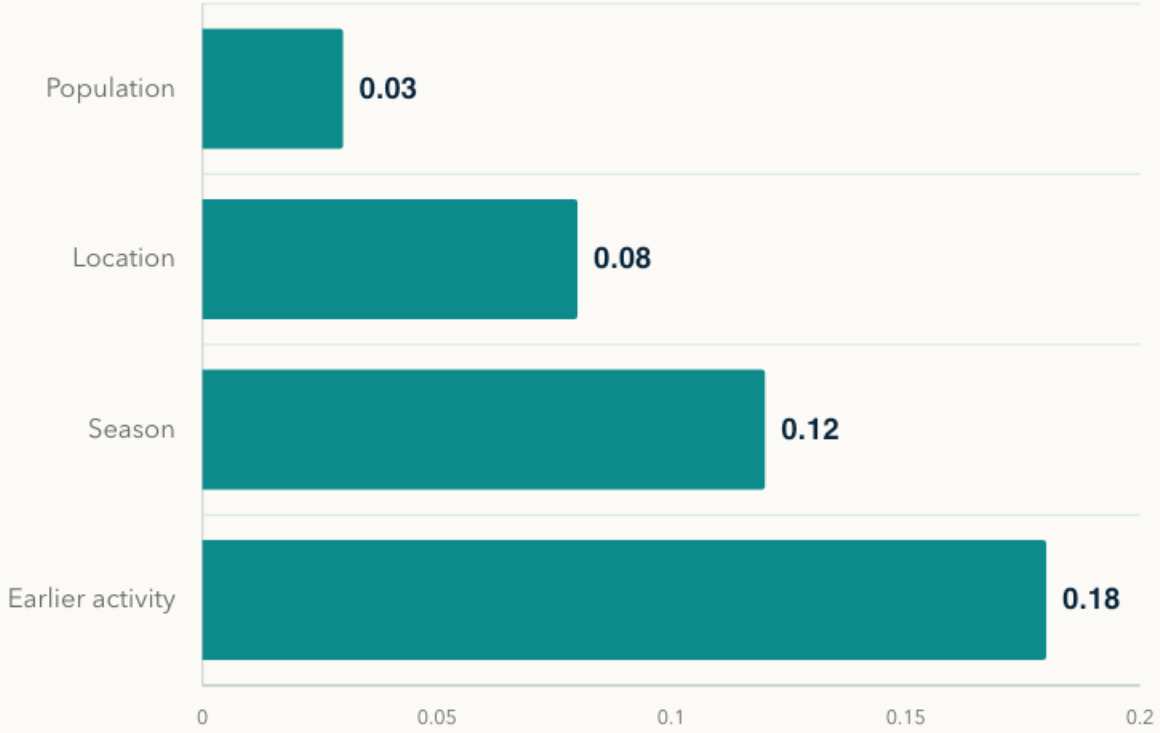
**Sensitivity = 32 / 40 = 80%**
**Specificity = 48 / 60 = 80%**




ROC shows sensitivity across thresholds. AUC summarises ranking performance, not the consequences of a chosen threshold.

# Variable importance and model interpretation

**Illustrative permutation importance**




## Permutation importance asks:

## How much does predictive performance worsen when one feature is disrupted?

- Use held-out data for performance-based importance
- Correlated features can share or mask importance
- Importance ranks predictive contribution, not causal influence

# Prediction versus explanation and causation

## Prediction

**Goal** **Estimate outcomes accurately for new observations**

**Success** Stable performance on unseen data

## Explanation and causation

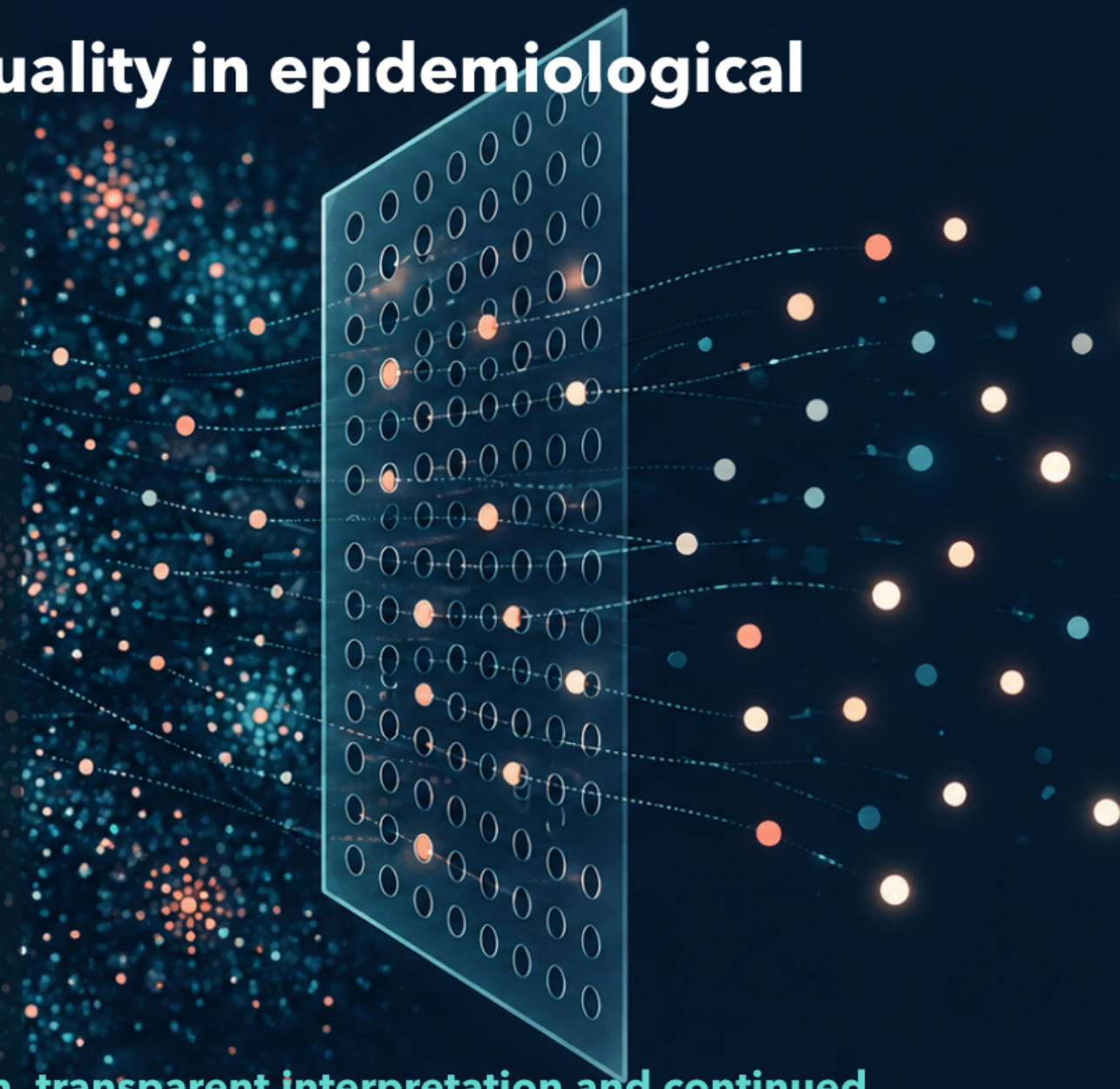
**Goal** **Understand a relationship or intervention effect**

**Success** A defensible design and assumptions for inference

**A highly predictive feature may be a proxy, a consequence of disease, or a product of the surveillance system.**

**Good prediction alone does not establish a cause.**

# Limitations, bias and data quality in epidemiological machine learning



## A technically correct pipeline can still produce misleading predictions when:

- Cases are missed or reported differently across place and time
- The training population differs from the population of use
- Missingness carries information that the model handles poorly
- Historical inequities become patterns the model repeats
- Performance drifts as surveillance and disease dynamics change

**Trust requires data checks, appropriate validation, transparent interpretation and continued monitoring.**